

REVIEW

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AI in power systems: a systematic review of key matters of concern

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Abstract

Recent advances in Artificial Intelligence (AI) have generated both excitement and concern within the power sector. While AI holds significant promise, enabling improved forecasting of renewable energy generation, enhanced grid resilience, and better supply-demand balancing, it also raises critical issues around transparency, data privacy, accountability, and fairness in power distribution. Despite the growing body of research on AI applications in power systems, there is a lack of structured understanding of the key socio-technical matters of concern (MCs) surrounding its integration. This paper addresses this gap by conducting a *systematic literature review combined with qualitative text analysis* to identify and synthesize the most prominent socio-technical concerns in the academic discourse. We analyzed a curated sample of peer-reviewed papers published between 1987 and 2024, focusing on high-impact journals in the field. Our analysis reveals four major categories of concern: (1) *Operational Concerns*-relating to AI's reliability, efficiency, and integration with existing grid systems; (2) *Sustainability Concerns*-centered on energy consumption, environmental impact, and AI's role in the energy transition; (3) *Trust Concerns*-including transparency, explainability, cybersecurity, and ethics; and (4) *Regulatory and Economic Concerns*-covering issues of accountability, regulatory compliance, and cost-effectiveness. By mapping these concerns into a cohesive analytical framework, this study contributes to the literature by offering a clearer understanding of AI's sociotechnical challenges in the power sector. The framework also informs future research and policymaking efforts aimed at the responsible and sustainable deployment of AI in power systems.

Keywords Concerns, Text analysis, Artificial Intelligence (AI), Power systems, Energy management

Introduction

As societies grapple with climate change, energy security, and environmental degradation, the transition from fossil fuels to sustainable clean energy power systems has become a critical priority. The rapid advancement of artificial intelligence (AI) has sparked both optimism and concern across various fields, including sustainable energy [1–3]. AI has been referred to as a coalescing force of the fourth industrial revolution [4]. As an assistive technology for the phenomenon of power system transformations

comprising both governance superstructures and technological infrastructures, AI affords many promising benefits [5–7]. AI has emerged as a powerful tool in the planning stage of power systems, offering solutions to operational challenges such as optimizing decision-making, forecasting power consumption through consumer usage pattern analysis [8], enabling smart load control [9], and automating the detection of anomalies in wind turbine components [10]. The integration of AI is evident across a wide range of power systems, from decentralized smart microgrids to large-scale centralized grids. Its significant role extends across various applications, including energy storage [11], power production [12], grid efficiency [13] and transmission [14].

However, the increasing integration of AI into power systems has also raised concerns among scholars, practitioners, policymakers, and consumers, prompting calls for further research on its broader implications [6, 11]. Key concerns include AI's ability to extract and cross-reference sensitive personal information from smart meters [15], biases embedded within AI algorithms [16], and risks of excessive surveillance and behavioral enforcement [17]. Additionally, evidence suggests that AI may exacerbate socio-economic disparities, particularly about equality, diversity, and inclusion [17–20]. Despite these concerns, research on AI in power systems remains fragmented. Existing studies often focus on either technical challenges, such as security and interoperability [21], or isolated social aspects, like user trust and AI explainability [22]. While these studies provide valuable insights, they fail to provide a structured and integrative framework to understand and connect the broader matters of concern (MCs) surrounding AI's integration in power systems. This fragmentation hinders a comprehensive understanding of AI's sociotechnical challenges and opportunities, limiting the development of holistic solutions. As the energy sector rapidly expands AI-driven transformations—an effort projected to reach \$13 billion according to the International Energy Agency (IEA) [23]—a systematic approach is needed to study AI's ethical, regulatory, and societal dimensions.

This study addresses this gap by systematically reviewing and analyzing the literature on AI in power systems and proposing a structured framework that maps related socio-technical concerns. Accordingly, this study seeks to answer two key research questions: *What are the primary concerns regarding AI's integration into power systems?* and *What specific contentions emerge from this integration?* To do so, text analysis is applied to examine the underlying meanings of these concerns. The analysis of these sociotechnical concerns aim to improve our understanding of the energy sector's likely changes with AI and help build a stronger theoretical framework, which can, in turn, guide the energy community in establishing future rules and policies for AI integration in power systems. Thus, this research aims to provide theoretical and policy insights to better understand AI's role in sociotechnical energy transformations [11, 17].

To accomplish these aims, the remainder of the paper is organized as follows: First, we present the theoretical foundation of sociotechnical transformations and Matters of Concern (MCs), establishing a framework for analyzing AI integration in power systems. Next, we outline the methodology used to identify and analyze key MCs. We then present our findings, categorizing them into four dominant MCs, each encompassing multiple Points of Contention (PCs). The discussion section follows, highlighting the most prevalent concerns, examining their interrelationships, and offering policy recommendations. Finally, the paper concludes with a summary of key insights and directions for future research.

Theoretics: sociotechnical matters of concerns

This interdisciplinary research is grounded in the fields of energy power systems and sociotechnical networks. It explores power system networks encompassing human interactions and technological advancements, alongside societal, cultural, political, and environmental influences [18]. Sociotechnical systems involve interlinked human, technological, and organizational elements, whose dynamic interactions collectively produce societal effects. Within this framework, agency emerges as a systemic feature of actor-networks, where the interplay between human actors and technological actants generates collaborative outcomes [24]. Matters of Concern (MCs) represent one of several theoretical frameworks situated within the broader Sociotechnical Systems (STS) literature, offering a structured approach to identifying and organizing complex, contested issues inherent in sociotechnical transformations [18, 25]. This makes MCs particularly suitable for analyzing AI integration into power systems, as they effectively capture the intricate interplay among technical, social, and political dimensions influencing technology adoption. MCs become particularly significant as they highlight nuanced contentions that both facilitate and constrain the formation and evolution of networks, such as the emerging AI-power systems configuration.

Matters of Concern (MCs)

The rise of AI as a technological infrastructure rapidly integrating into the everyday work and play of human networks is widely recognized as a matter of fact. However, the superstructural challenges associated with this integration—such as how humans will govern, set boundaries for, and incorporate AI into complex power systems actor-networks—are less understood [26]. This paper proposes that to grasp these critical superstructural aspects, research must focus on the energy community's MCs regarding the integration of AI.

While extensive research has explored the individual knowledge bases underlying AI integration into power systems—such as unresolved privacy issues related to smart meter data—the connections between these discrete concerns and their broader implications have received less attention [27]. Concerns about AI in power systems are diverse and multifaceted. For instance, consumer concerns regarding privacy and interoperability [28] are closely tied to regulatory compliance issues from a governmental perspective [29]. Addressing these interconnected concerns is essential for developing a comprehensive understanding of AI's role in power systems.

Studies in other fields, such as the development of social robots, have successfully employed MCs to identify and address ethical issues for senior citizens [30]. Other research has shown how MCs can be a useful theoretical tool to identify inadequacies within supply chain relationships along with management accounting, going against the conventional wisdom applied in both areas [31]. Even on contentious issues such as sustainable meat, MCs can construct an understanding through a qualitative analysis of public discourse on newer social media platforms [32]. More importantly, for similarly complex grand challenges, researchers have studied MCs in relation to the emergence of assistive AI in creativity [33] as well as MCs arising from the difficult and dangerous challenge of commercializing outer space [34].

The concept of MCs is particularly useful for identifying specific sociotechnical situations that spark public discourse due to epistemological and ontological uncertainties

[35]. These uncertainties often stem from limited knowledge, leading to subjective and sometimes contentious feelings. MCs persist because there is no consensus or overarching meta-agreement on them, and they are maintained by actors deeply invested in these concerns. Moreover, MCs become evident when matters of fact are considered within their full contextual frameworks [35]. For instance, incomplete facts evolve into MCs when they are integrated into sociotechnical applications that meaningfully connect human lives with emerging technologies, often with uncertain consequences for those lives.

From a theoretical standpoint, MCs offer both evidence and underlying cultural meanings of sociotechnical systems. As revealed through contentions and debates, MCs expose sentiments, assumptions, and insights about human responses and acceptance [36–38]. Therefore, identifying and analyzing MCs is crucial for understanding the development of AI and power systems and the extent to which stakeholder expectations, values, and assumptions conflict or align, as expressed in points of contention [39]. MCs become apparent through the discourse and sentiments expressed by actors involved in a particular phenomenon [35]. These concerns encompass topics that reveal underlying PCs, which coalesce into MCs, that evoke strong emotions and thereby indicate levels of acceptance or resistance to emerging sociotechnical systems. In this study, the primary interest lies in understanding why and how MCs and their PCs surrounding AI, power systems, and governance emerge, take shape, and inform theoretical conceptualization [40]. In the context of AI and power systems, MCs offer a valuable framework for understanding these concerns, as they do not fit neatly into a single concept. Instead, they form an encompassing network that allows for the classification and examination of these complex concerns across multiple dimensions. This comprehensive approach enables a more nuanced investigation of emerging sentiments and prediction of future developments for AI-power system networks.

Method

Research approach

To contribute to the growing discourse on AI in power systems and expand existing theoretical frameworks, this study employs an exploratory qualitative research design tailored to its aims and questions, utilizing textual analysis of discourse technique [41–43]. This interdisciplinary approach bridges context and theory phenomenologically, offering deeper insights into MCs and PCs. By leveraging exploratory text analysis, the study identifies dominant discourses, debates, and contentions within a carefully curated collection of scholarly literature, ensuring a focused and meaningful examination of the phenomenon under investigation [44].

Data collection: corpus

We curated a focused corpus to facilitate a thorough and nuanced exploration of key issues, themes, and debates within a well-defined context. The corpus ($N_{CASES} = 63$) consists of peer-reviewed journal articles selected from Scopus, IEEE Xplore, ScienceDirect, and Google Scholar, based on their relevance to AI in power systems, publication in English, and journal quality (restricted to first- and second-quartile journals). To identify relevant publications, we employed a combinatorial search strategy, systematically pairing terms from two broad categories—**AI techniques** and **energy systems** (e.g., "*machine*

learning" AND *"power systems"*, *"deep learning*" AND *"electricity grid"*"), and refining the results to those most relevant to the intersection of AI and power systems.

The individual keywords used from the two conceptual categories were as follows: for **AI techniques** (e.g., *"machine learning"*, *"deep learning"*, *"expert systems"*, *"neural networks"*, *"AI"*, *"intelligent systems"*, *"multi-agent systems"*, *"agent-based models"*, *"artificial intelligence"*, *"internet of things"*) and for **energy systems** (e.g., *"power systems"*, *"electricity grid"*, *"smart grid"*, *"power grid"*, *"energy system"*, *"renewable integration"*, *"demand-side management"*, *"microgrid"*, *"energy forecasting"*, *"grid optimization"*, *"renewable energy"*, *"solar forecasting"*, *"predictive maintenance"*).

While not a systematic review in the formal sense, this strategy followed a structured and replicable process, informed by the authors' domain expertise and tailored to the scope and purpose of the discourse analysis.

We excluded non-academic articles, opinion pieces, and papers that did not directly address AI in power systems. Although AI applications in power systems gained heightened prominence post-2010, their origins trace back several decades, reflecting a sustained and evolving research trajectory. As detailed by Heymann et al. [45], AI methodologies-including expert systems and early neural networks-have been actively explored within power systems since at least the early 1980 s. To provide a comprehensive historical perspective and capture foundational developments, our analysis thus spans from 1987 to 2024, encompassing the evolution of AI terminology and methodologies in the power sector-from early expert systems [46] to contemporary approaches such as neural networks [47], machine learning [48], smart grids, and autonomous power systems [49]. The inclusion of AI-related synonyms contextualizes technological advancements across different periods. Each document was compiled into a structured dataset and processed using a readtext package in R to extract textual content frequencies. Metadata, including authorship and publication year, were derived from filenames and integrated into the corpus for further analysis. The readtext package provides a user-friendly way to import and handle text data from various file formats including PDFs. It creates a data frame containing the document text and metadata, simplifying the corpus creation process. The package automatically extracts text content while preserving document identifiers, which facilitates further text mining operations. For more details, readers can refer to Benoit and Obeng [50].

Research process: stages

Our research followed an iterative process that combined human close reading of the collected corpus, AI-assisted validation, and computational text analysis to systematically identify discourse patterns within the literature on AI in power systems. The goal was to develop a holistic, system-level perspective on AI integration by uncovering key debates, concerns, and their interconnections, thereby exposing broader sociotechnical challenges. Our multi-stage iterative approach analyses progressed recursively between stages rather than linearly, facilitating continuous refinement of thematic categorizations. Each iteration involved revisiting previous stages to reassess and update thematic classifications based on emerging insights. This iterative process continued systematically until thematic saturation was achieved-defined as the point at which further iterations no longer yielded additional themes or modifications. Consequently, the final dictionary reliably encapsulated a comprehensive set of key thematic areas relevant to

artificial intelligence discourse within power systems research. The main stages are as follows:

Initial Identification of Debates: We employed a mixed-method approach to reviewing the literature. The initial stage involved a systematic close reading of a portion of the corpus, focusing on sections that explicitly highlighted concerns, key debates, and root keywords. These debates may be expressed as fully developed Points of Contention (PCs) with associated keywords and topics - lines of reasoning that contribute to a broader argument. We manually extracted and documented relevant concerns, debates, discussion topics, and root keywords.

Deriving Initial Matters of Concern (MCs): Matters of Concern emerged as higher level, strategic groupings of multiple interrelated topics and PCs. These were only discernible after systematically analyzing and manually organizing the initial PCs with topics above into thematic categories. To illustrate, larger concern themes, such as “Trust Concerns” emerged from our team discussing and manually grouping together, due to perceived interrelatedness, the PCs, “Explainability” (comprising topics such as “Multi-energy systems” and “Explainability’s role”) and “Data Governance and Security” (comprising topics such as “Data availability and standardization” and “Data quality and bias”).

Validation and Expansion Using Generative AI: To enhance comprehensiveness and mitigate potential bias, we employed generative AI models, Gemini Advanced and ChatGPT-3.5, to analyze the full corpus. These tools were prompted to identify additional debates, topics, and sub-themes potentially overlooked during manual review. Prompts included: “Please provide more detail for each of these new matters of concern” and “Please provide more specificity and sub-themes for each key matter of concern.” This allowed us to triangulate and extend the preliminary coding structure.

Dictionary construction

We then expanded the initial list of root keywords into a comprehensive custom dictionary ($N_{KWDs} = 891$). We used ChatGPT-3.5 to validate the existing terms and suggest additional synonyms and semantically related terms. The keywords act as semantic markers that allow for deeper computational analysis, i.e., to facilitate the automated identification and quantification of the debates above across all the documents [51]. Prompts such as “For each key matter of concern and sub-theme, please provide keywords that I might use to search for other similar literature” guided this phase. Importantly, we did not rely on publicly available dictionaries to maintain contextual fidelity with our curated corpus. The resulting keyword sets were mapped hierarchically to specific topics, PCs, and MCs. For example, under the topic “Accuracy and Reliability,” keywords included terms like “forecast accuracy,” “grid imbalances,” “conservative forecasts,” “underutilize renewable,” and “grid reliability.” We concluded the dictionary by using expert validation through collaborative review sessions with our team, including a domain specialist, to ensure keyword relevance and thematic alignment based on sample text reexamination.

Computational text analysis

Once finalized, the dictionary was applied to the full corpus using the Quanteda package in R for text mining, term frequency (TF), and co-occurrence analysis. The dictionary

enabled tokenization of the text, matching of terms (including bigrams and trigrams), and analysis of semantic associations between themes. TF was selected as the primary metric due to its proven value in domain-specific corpora [52, 53], with higher TF values indicating greater relative prominence in the literature [54].

Our calculation involved the following steps:

1. Text tokenization with removal of punctuation, numbers, and stopwords;
2. Generation of n-grams (2–3 word combinations) to preserve multi-word technical terms;
3. Compounding of recognized multi-word expressions;
4. Dictionary lookup against cleaned and tokenized text.

This ensured that domain-specific terms were counted holistically rather than as disaggregated tokens. To evaluate relationships between categories, we conducted co-occurrence analysis (i.e., how often two themes appeared in the same document) and calculated the Jaccard Coefficient (JC) to quantify the degree of overlap between term sets—highlighting which MCs were frequently debated in tandem versus treated independently [55]. The results from this process are detailed in the next section.

Results

This section presents the findings from our text analysis of the corpus literature focusing on identifying the MCs, PCs, and topics of discussion surrounding the integration of AI in power systems, their interrelations, and broader implications. Given the complexity and nature of sociotechnical debates, MCs, PCs, and topics overlap; we have positioned each element where it is most thematically prominent, while their interconnections highlight the nuanced, multifaceted nature of the discourses. Table 1 shows the four dominant MCs that we identified from the literature, sorted by their term frequencies (TF). The subsections below describe each MC in greater detail.

Operational concerns: AI as grid coordinator

AI's growing role in managing and optimizing power grids with high renewable energy penetration has sparked debate, making it the most frequently discussed Matter of Concern (MC) in our corpus (TF = 22,083; NC = 63; 100%). The integration of wind and solar energy, for example, introduces technical challenges due to output variability, forecasting uncertainties, and the need for rapid system adjustments. While AI enhances real-time control, energy distribution, and grid resilience, its black-box nature, reliance on large-scale data, and interaction with legacy infrastructure pose risks.

Advocates argue that AI can serve as a valuable grid coordinator, dynamically adapting to new energy sources, shifting demand patterns, and evolving regulations. However, others question its reliability, warning that opaque decision-making, unexpected system behaviors, and coordination difficulties with traditional grid protocols may introduce

Table 1 Key Matters of Concern (MC)

MC	TF	N Cases	%N Cases
<i>Operational Concerns</i>	22,083	63	100
<i>Sustainability Concerns</i>	11,940	63	100
<i>Trust Concerns</i>	7,629	63	100
<i>Regulatory and Economic Concerns</i>	3,831	61	97

new disruptions rather than solving existing ones. Addressing these concerns is critical to ensuring AI-driven grid management remains reliable, secure, and transparent.

For example, Ahmad et al. [48] emphasize AI's ability to improve efficiency, energy management, and transparency in renewable power systems:

"Artificial intelligence (AI) is starting to play a major role in the energy market... AI technologies will lead to the improvement of efficiency, energy management, transparency, and the usage of renewable energies. In recent decades/years, new AI technology has brought significant improvements to how power system devices monitor data, communicate with the system, analyze input-output, and display data in unprecedented ways. New applications in the energy system become feasible when these new AI developments are incorporated into the energy industry." (p. 334)

Trivedi and Kadam [56] similarly highlight AI's potential in microgrid environments, particularly when integrated with hierarchical control schemes:

"Adjoining AI techniques with these existing schemes can bring higher accuracy, speed, and better effectiveness for control and operation in a microgrid environment." (p. 1)

Such hierarchical control schemes help manage power system operations at different levels, including voltage and current regulation, power flow optimization, synchronization, and economic dispatch [57], reinforcing AI's role in grid modernization.

Despite these advantages, serious concerns remain about AI's practicality and feasibility in large-scale power systems. Liu et al. [58] warn that outdated control infrastructure limits AI adoption, as legacy grids lack the flexibility to support advanced AI-driven mechanisms:

"The existing control equipment makes it difficult to support the new renewable energy generation network. Therefore, these old control devices need to be updated to support the new layout of the energy sectors." (p. 15)

Beyond infrastructure limitations, the complexity and opacity of AI models introduce risks for grid stability. Machlev et al. [22] highlight how AI's lack of explainability and decision transparency can disrupt grid management, as operators may struggle to align AI-driven actions with established operational standards:

"Due to the growing penetration of renewable sources, liberalization of electricity markets, and increasing presence of direct current transmission systems, the power grid's vulnerability is increasing. Recently, advanced AI techniques have shown great promise in dealing with stability analysis and other control-related aspects, but the practical implementation of these AI techniques has been hindered by the lack of transparency of these black-box models... This lack of interpretability makes it difficult for system operators to diagnose AI-driven anomalies and take corrective action before disruptions occur." (p. 5) [22]

This mismatch between AI strategies and legacy infrastructure raises doubts about AI's real-world applicability. The challenge of scaling AI solutions from small pilot projects to fully integrated grid operations remains significant. Ahmad et al. [13] also caution that

existing communication protocols may not support AI-driven automation, further complicating integration:

“The synchronization of current equipment with new equipment produces some smart production technology deployment challenges. At this time, the old protocol is difficult to support the communication of the new device, it needs a new protocol to replace old protocols to support new machines in the energy sector.” (p. 352)

As a result, two major points of contention emerged in our analysis:

1. AI's *Instrumentality* - How effectively it can manage renewable energy variability.
2. AI's *Integrability* - Whether it can seamlessly integrate into current power grid infrastructure.

These challenges are central to evaluating AI's role as a grid manager. Table 2 illustrates the relationship between this MC and its two primary Points of Contention (PCs). The prevalence profiles include term frequencies (TFs), the number of cases, and the percentage of cases for each PC, highlighting the significance of AI's control capabilities and integration challenges in power systems.

Instrumentality PC (TF = 20, 507; 63 cases; 100%)

This PC focuses on technical debates regarding AI's ability to enhance renewable power systems by improving forecasting accuracy, managing intermittency, and addressing methodological trade-offs. It examines AI's role in advancing specific tools and innovations, distinct from broader discussions on balancing competing interests. Main topics include:

- *Accuracy and Reliability* (12, 924 TF, 60 cases, 95%): AI's ability to improve energy generation forecasts is crucial, but weather variability introduces significant uncertainty. Researchers are refining AI models to handle outliers and anomalies in weather data to prevent grid imbalances [59]. While Trivedi and Kadam [56] emphasize AI's potential for precise control, others question whether AI can consistently deliver reliable predictions. The contention centers on balancing forecast precision with the risk of under- or over-utilizing renewable resources [6]. For instance, Gupta et al. [60] argue that AI models should be assessed on their ability to process and filter corrupt or incomplete data inputs.
- *Methodological Choices and Trade-offs* (6, 479 TF, 63 cases, 100%): Various AI techniques, from traditional methods like support vector machines to advanced deep

Table 2 Key PCs and topics under operational concerns: AI as grid coordinator

Operational Concerns: AI as Grid Coordinator	TF	N Cases	% N Cases
PC: Instrumentality	20507	63	100
Accuracy and reliability	12924	60	95
Methodological choices and trade-offs	6479	63	100
Handling intermittency	1104	45	71
PC: Integrability	1576	59	94
Scalability and interoperability	744	57	90
Stability vs. Destabilization	640	41	65
Compatibility with legacy systems	98	19	30
Managing real-time data	76	16	25
Adaptation to evolving energy landscape	18	7	11

learning approaches, such as convolutional neural networks, offer different levels of accuracy, complexity, and computational demands [61]. Simpler models provide faster results but lower precision, especially for long-term forecasts, while complex models increase accuracy at a higher computational cost. For example, support vector machines are efficient for real-time grid management but may struggle with complex, nonlinear weather patterns. In contrast, convolutional neural networks offer superior forecasting accuracy but require more computational resources [47]. The debate revolves around optimizing AI for different time horizons in energy management.

- *Handling Intermittency* (1, 104 TF, 45 cases, 71%): AI is increasingly seen as a tool to manage the anticipated and known intermittency of renewable energy sources by predicting periods of high or low generation and adjusting grid operations accordingly [17]. However, technical capabilities for managing real-time fluctuations are still developing. The challenge is whether AI can evolve fast enough to meet the demands of large-scale, real-time energy grids and effectively smooth out the inherent variability in renewable energy production [62, 63]. For AI to succeed, it must integrate diverse data sources—such as weather forecasts, historical consumption patterns, and real-time sensor data—to create a holistic view of the energy landscape [64].

Integrability PC (TF = 1, 576; 59 cases; 94 %)

This point of contention examines the technical challenges of integrating AI into large-scale power systems, particularly those reliant on legacy infrastructure. Key concerns include scalability, interoperability, system stability, compatibility with older technologies, and real-time data management in an evolving energy landscape. These challenges impact grid operators, regulators, and end-users, all of whom depend on a stable and interoperable network. Key topics include:

- *Scalability and Interoperability* (744 TF, 57 cases, 90%): Scaling AI from controlled pilot projects to fully integrated power grid systems across diverse regions is challenging. Interoperability, or ensuring AI's seamless operation across various vendor systems with different communication protocols, is a major contention. The debate revolves around whether AI can maintain performance at scale without being hindered by compatibility issues across heterogeneous networks [58].
- *Stability vs. Destabilization* (640 TF, 41 cases, 65%): AI-managed power grids risk destabilization if control decisions are made too quickly or without sufficient oversight, potentially causing oscillations or voltage fluctuations that compromise grid stability [65]. While AI systems undergo rigorous testing in both simulations and real-world settings, debates continue over whether they can reliably maintain the delicate balance required for grid stability [66].
- *Compatibility with Legacy Systems* (98 TF, 19 cases, 30%): This topic captures the technical difficulties of integrating AI with aging physical and digital infrastructure. Many existing power grids were not designed to support the data processing, communication protocols, or interoperability demands of contemporary AI applications. Integration often requires costly retrofitting or replacement of outdated components, raising concerns about feasibility and return on investment [1].

- *Managing Real-Time Data* (76 TF, 16 cases, 25%): AI must efficiently process vast streams of real-time data to manage energy distribution and grid operations. This involves significant processing resources and the use of advanced algorithms capable of handling large data volumes under strict time constraints. The debate centers on whether current AI systems have the computational power to process real-time data without delays or bottlenecks [61], as well as solve complex NP-hard optimization problems related to power grid operation (i.e., power generation, transmission, and distribution) [67].
- *Adaptation to the Evolving Energy Landscape* (18 TF, 7 cases, 11%): As discussed before, the energy sector is undergoing rapid transformation, driven by the integration of new renewable energy sources, demand-response strategies that shift consumption patterns, and evolving regulatory frameworks. AI systems must adapt in real-time to keep pace with these dynamic changes. This topic of discussion focuses on the long-term adaptability and responsiveness of AI to external pressures and structural shifts. A key concern is whether AI can evolve quickly enough-by re-training algorithms with updated datasets-to address these changes without causing disruptions, thereby ensuring its long-term sustainability in the energy sector [63, 68].

Sustainability concerns: AI for sustainable energy

The Sustainability MC examines AI's role in enhancing the societal and environmental sustainability of power systems. It ranks second in prevalence, with a term frequency (TF) of 11,940 across all 63 cases (100%). The debate surrounding AI's role in sustainable energy revolves around its *adoptability*-how readily it can be integrated into power systems-and its *adaptability*-its ability to remain effective as energy demands and environmental conditions evolve.

Proponents argue that AI can accelerate renewable energy deployment and optimize energy use, both of which are essential for achieving climate goals. They highlight AI's potential to enhance real-time energy management, forecasting accuracy, and grid efficiency, enabling a more sustainable transition to clean energy sources. However, others raise concerns about AI's own energy consumption and environmental footprint, particularly the high computational demands associated with training and running complex models [69]. Additionally, retrofitting AI into existing power infrastructure presents challenges, as many legacy systems are not designed for AI-driven automation, creating compatibility and scalability issues [70].

While supporters advocate for faster adoption to maximize AI's sustainability benefits, they acknowledge the need for strong policies and integration frameworks. Others argue that AI should function as a tool for balancing multiple objectives, such as reducing environmental impact, maintaining user comfort, and increasing energy efficiency [71]. At the same time, concerns remain about whether AI prioritizes efficiency at the expense of equity, user autonomy, or resource sustainability. Ahmad et al. [69] highlight these concerns:

"While AI development is an effective and promising development of sustainability, implementation produces an enormous amount of carbon footprint, reflecting a direct rebound effect. A single AI learning algorithm can release CO₂ emissions equal

to five cars. AI techniques rely heavily on different types of energy data, thus contributing indirectly/directly to the global carbon footprint of information technology.” (p. 8) [69].

Another major concern is AI’s influence on consumer behavior [17]. AI-powered energy management systems can encourage more sustainable energy use by providing personalized recommendations and demand-response incentives. However, critics question the ethical implications of AI-driven behavioral modification [72]. Some argue that AI might prioritize system efficiency over user autonomy, potentially incentivizing energy use patterns that create unintended risks [73]. Privacy concerns and the potential for data misuse and manipulation raise further doubts about AI’s ability to genuinely foster sustainability [74]. Godoy et al. [75] warn of these broader risks:

“We can say that the monitoring of people’s behavior and the transgression of their privacy rights is identified as a threat to trustworthy cultures because their democratic processes and practices, central to their continuous development and welfare, are challenged by the misuse of data for inference and modification of users’ behavior and preferences, driven by corporate or governmental interests.” (p. 8) [75].

Thus, the Sustainability MC highlights the tension between AI’s potential to advance environmental goals and the practical and ethical challenges of its implementation. While supporters view AI as a driver of sustainable energy management, critics argue that its energy footprint, influence on consumer behavior, and ethical risks may undermine its ability to truly support a sustainable future. Table 3 presents the prevalence profiles for AI’s Sustainability MC and its two primary Points of Contention (PCs), *Adoptability* and *Adaptability*, along with their associated topics.

Adoptability PC (TF = 10, 366; 63 cases; 100%)

The Adoptability PC explores the overarching “why” behind AI adoption, focusing on systemic readiness and stakeholder willingness to implement AI at scale. Unlike purely technical considerations, this PC examines the broader factors that influence AI’s acceptance and integration within the energy sector. Key debates center on the rate and ease of adoption, particularly in the context of urgent environmental goals. This includes discussions on barriers to implementation, ethical concerns, and the level of preparedness among stakeholders to incorporate AI into routine operations. Main topics of discussion include:

Table 3 Key PCs and topics under sustainability concerns: AI for sustainable energy

Sustainability Concerns: AI for Sustainable Energy	TF	N Cases	% N Cases
PC: Adoptability	10366	63	100
Acceleration potential	4019	62	98
Industrial process optimization	2236	59	94
Behavior change	1432	57	90
Balancing comfort and savings	1419	53	84
Holistic approaches	1260	53	84
PC: Adaptability	1574	60	95
Energy consumption	1350	54	86
Performance trade-offs	224	42	67

- *Acceleration Potential* (TF = 4, 019; 62 cases; 98%): AI is often heralded as a tool for accelerating the deployment of renewable energy technologies and optimizing demand-side management. However, the speed of AI adoption remains a significant concern. Although AI can theoretically optimize power systems quickly, real-world implementation faces regulatory and technical barriers, such as outdated superstructures and infrastructures, which impede its immediate impact on sustainability [76].
- *Industrial Process Optimization* (TF = 2, 236; 59 cases; 94%): the AI and energy literature discuss the social and technical challenges involved in retrofitting industrial processes with AI. This primarily impacts entities like prosumers, industrial operators, and even large-scale consumers who are directly involved in energy production or high energy use. These stakeholders would need to strategically consider the challenges and benefits of adopting AI to optimize their processes, where AI could improve efficiency but would require significant investments and adjustments in superstructures and infrastructures, training, and ongoing support. For instance, integrating AI in energy-intensive industries requires balancing potential efficiency gains against substantial investments in technology, training, and ongoing operation and maintenance costs [66].
- *Behavior Change* (TF = 1, 432; 57 cases; 90%): AI's role in influencing sustainable consumer behavior is widely debated. Proponents argue that AI can encourage energy-saving practices by offering real-time feedback and personalized recommendations. However, concerns about the moral implications of AI-driven behavior modification, particularly regarding privacy and consent, raise questions about the appropriate extent of AI's influence on individual choices in the name of sustainability and the users' willingness to adopt the technology [74, 75].
- *Balancing Comfort and Savings* (TF = 1, 419; 53 cases; 84%): A key challenge in AI-driven power systems is the need to balance energy efficiency with user comfort. While AI can optimize energy savings, it may do so at the expense of convenience, such as by regulating thermostats or programming appliances to operate during off-peak hours. This tension between sustainability goals and user satisfaction presents a challenge for AI's widespread adoption in residential and commercial settings. Critics advocate for a multi-objective approach that balances optimal scheduling with cost savings, without compromising user comfort [68].
- *Holistic Approaches* (TF = 1, 260; 53 cases; 84%): While AI is a powerful tool for optimizing specific aspects of power systems, it cannot serve as a standalone solution. Achieving long-term sustainability requires a comprehensive, multi-faceted strategy that integrates technological advancements, behavioral changes, and supportive policies. AI should function as a complementary tool rather than a replacement for broader sustainability initiatives. This underscores the need for an integrated approach that extends beyond AI's immediate capabilities to address the complex challenges of energy sustainability [19].

Adaptability PC (TF = 1, 574; 60 cases; 95%)

The Adaptability PC explores AI's ability to remain effective as energy demands and environmental conditions evolve while also considering its own environmental footprint as a high-energy-consuming technology. The debate centers on whether AI can sustain

its performance as power systems become increasingly complex and dynamic-especially when these changes are not fully captured in the data used to train AI models-and on the significant power demands of AI data centers. Key topics within this contention include:

- *Energy Consumption* (TF = 1, 350; 54 cases; 86%): AI's energy consumption, particularly in large-scale models, continues to raise concerns about its overall environmental impact. The process of training and operating complex AI-based models is often energy-intensive, potentially offsetting some of the benefits. Proponents argue that advancements in efficient AI design and hardware could mitigate these effects. However, questions persist regarding whether such innovations can keep pace with the growing computational demands of AI, especially given projections of a steep increase in data centers' share of global electricity consumption [77].
- *Performance Trade-offs* (TF = 224; 42 cases; 67%): Achieving high levels of accuracy in AI systems frequently results in increased computational complexity, leading to a trade-off between performance and energy consumption. This challenge is particularly relevant in real-time applications, such as power grid management, where AI must rapidly process vast datasets to make timely decisions and adjustments. While high-performance models offer greater precision, they also require substantial resources to operate and train, raising contentions about the long-term sustainability of such systems [78].

Trust concerns: AI as a transparent and ethical agent

The Trust MC examines concerns surrounding AI's role in power systems, focusing on its explainability or transparency, data governance, and security. It ranks as the third most prevalent MC, with a term frequency (TF) of 7,629 across all 63 cases (100%). The increasing influence and integration of AI into energy management raise critical questions about the risks associated with opaque decision-making, user trust, and security vulnerabilities [79]. The debate revolves around whether AI can operate as a trustworthy energy agent, making decisions that are both explainable and ethically sound while minimizing risks to users, infrastructure, and society at large [6].

Proponents argue that AI can make power systems more efficient by improving real-time decision-making, error detection, and system monitoring. AI-driven analytics and machine learning models are believed to provide faster, more accurate insights than traditional methods, leading to better risk mitigation and improved system resilience. For instance, Mohammad and Mahjabeen [63] highlight how AI can proactively detect faults in solar power systems, increasing reliability and reducing downtime:

“Numerous operational problems and component breakdowns can impede the production of electricity in solar power systems. Machine learning techniques are used by AI-based fault detection and maintenance systems to assess real-time data from sensors and monitoring equipment installed inside the solar energy system. AI algorithms can rapidly find and diagnose problems by locating abnormalities, variances, or possible defects. As a result, preventive maintenance and repairs are possible, reducing system downtime and increasing energy output. Furthermore, AI algorithms can reveal the underlying causes of errors, assisting system administra-

tors in the implementation of preventative measures and enhancing overall system reliability.” (p. 119)

However, critics warn of the “**black box**” problem in AI-driven decision-making, where the lack of explainability undermines trust in its outputs [22]. Typically, AI’s recommendations and automated actions lack transparency and are based on thousands of factors, making it difficult for operators and decision-makers to make sense and assess how decisions are made or even intervene when necessary [80]. This opacity raises concerns about AI’s accountability, fairness, and biases, as, for example, system failures or unintended consequences may not be easily trackable or have a clear chain of responsibility.

“One of the main challenges is how to use models that have both high performance and transparency. Generally speaking, accurate models are more complex and harder to understand. This trade-off is particularly important in power systems, where operators demand both high accuracy and explainability to ensure system reliability. The lack of clear definitions and standardization in explainability further complicates AI adoption, as different stakeholders-AI researchers, power experts, and policymakers-have varying levels of abstraction in their decision-making processes” (p. 7) [22]

Concerns also extend to cybersecurity and data privacy risks. As AI becomes more deeply embedded in power systems, it increases exposure to cyber threats that could compromise energy security. AI-enabled smart grids rely on vast amounts of data, making them susceptible to breaches and manipulations that could disrupt critical infrastructure [81]. Farhan et al. [82] highlight these risks:

“The inclusion of smart grids and power systems into IoE further increases the danger of cyberattacks directed at crucial infrastructure. These systems are vulnerable to breaches that might cause power outages, financial losses, and even safety risks. The integrated internet-based smart grid and energy resources inherit all the security flaws of the current smart grid. The smart grid’s security framework is no longer sufficient to fulfill the security requirements of the energy sectors.” (p. 17)

Thus, this MC highlights the trade-off between AI’s potential benefits and the risks of opacity, security vulnerabilities, and ethical dilemmas. While AI has the capacity to improve efficiency and decision-making, ensuring its transparency, explainability, and ethical compliance is crucial for maintaining public and industry trust. The following sections examine these concerns through their points of contention *Explainability* and *Data Governance and Security*, illustrated in Table 4.

Explainability PC (TF = 5, 758; 62 cases; 98%)

The Explainability PC refers to the challenges of interpreting the rules that drive AI’s actions and decisions. Trust in AI’s decision-making is crucial for its effective adoption and acceptance in these systems. However, many AI models, especially those based on deep learning, function as “black boxes,” with their internal processes remaining opaque. The key topics identified in our analysis related to this contention are:

- *Multi-power systems* (TF = 3, 202; 55 cases; 87%): This topic addresses the challenges of tracking, understanding, and interpreting AI’s decision-making processes in

Table 4 Key PCs and topics under trust concerns: AI as a transparent and ethical agent

Trust Concerns: AI as a Transparent and Ethical Agent	TF	N Cases	% N Cases
PC: Explainability	5758	62	98
Multi-power systems	3202	55	87
Interpretability	2206	60	95
Black box problem	161	26	41
Model uncertainty	82	25	40
Maintaining Resilience	58	16	25
Deployment in real-world conditions	48	15	24
Conflict with traditional control systems	1	1	2
PC: Data Governance and Security	1871	58	92
Data availability and standardization	818	51	81
Data quality and bias	464	44	70
Data privacy and security	190	17	27
Cybersecurity vulnerabilities	178	15	24
Consumer data ownership	162	18	31
Information monopolization	59	22	35

complex power systems. AI faces significant difficulties in integrating and optimizing both current and future power grids, as multiple interconnected elements contribute to multi-directional energy flows-not just from producers to consumers. Future power grids will incorporate multiple and variable energy sources, battery storage systems, prosumers, electric vehicles, and mini-grids, among others. These diverse, multidirectional energy flows add layers of complexity beyond existing operational norms, requiring advanced coordination capabilities. As AI manages these intricate interactions, stakeholders will need transparency to understand how AI allocates resources, maintains grid stability, remunerates services, and optimizes energy flows in real-time. Without clear explanations of AI's decision-making, producers and operators may struggle to trust or effectively oversee these advanced systems, increasing the risk of inefficiencies or grid imbalances [62].

- *Interpretability* (TF = 2, 206; 60 cases; 95%): Interpretability refers to the degree to which humans can cognitively understand the logic behind how AI models reach decisions. While such models may process and combine hundreds of thousands of variables to generate an outcome, the underlying reasoning is not always immediately clear, even when the steps and parameters are technically accessible. In this context, the challenge lies not in accessing the process itself, but in the cognitive difficulty of making sense of it. This topic captures how effectively an AI system can present its decision-making process in a way that is interpretable and usable, which is essential for fostering trust in AI-driven power systems. Interpretability enables operators to trace decisions, adjust processes, and prevent unintended interactions. A key challenge, therefore, is the trade-off between model performance and interpretability: more complex and accurate models, such as deep neural networks, often become increasingly opaque. The absence of interpretability may hinder the adoption and oversight of AI in critical infrastructure [56, 81].
- *Black Box Problem* (TF = 161; 26 cases; 41%): As mentioned above, AI models often make decisions through complex interactions and rules involving thousands, or even millions, of parameters, making it virtually impossible to clearly lay out the reasoning sequence behind a specific output. The black box problem refers to this lack of transparency in advanced AI models, where the sheer scale and complexity

of the decision-making process render the steps practically impossible to reconstruct or inspect. This opacity presents major challenges for validation, accountability, and oversight, especially in critical domains like power systems. Without the ability to clearly demonstrate and trace the logic behind AI outputs, stakeholders struggle to diagnose errors, ensure fairness, or intervene in real time, ultimately weakening trust and limiting the safe integration of AI into decision-making processes [19, 22].

- *Model Uncertainty* (TF = 82; 25 cases; 40%): AI models often face uncertainty in their outputs, especially when dealing with unpredictable variables as inputs such as weather conditions or shifting consumer behavior. These models may struggle to make accurate predictions when confronted with unforeseen and unanticipated events, such as those driven by climate change. This uncertainty can undermine confidence in AI's decisions since accurate forecasting and public trust in those forecasts are vital for effective system coordination [7, 58, 68].
- *Maintaining Resilience* (TF = 58; 16 cases; 25%): In this topic, scholars highlight the need for AI to balance efficiency with resilience, ensuring that power systems can withstand and recover from disruptions. Resilience is especially critical for systems dependent on variable renewable energy sources. Despite potential inefficiencies, AI models must be designed to incorporate redundancy and maintain operational stability, preventing single points of failure that could compromise system reliability [1, 18].
- *Deployment in Real-World Conditions* (TF = 48; 15 cases; 24%): AI models are often developed and tested in controlled environments, which may not fully reflect the empirical complexity of in situ power systems. When deployed, these models may face unforeseen conditions beyond their training. So, this topic discusses the adaptability of AI models when moving from controlled development settings to unpredictable real-world environments. The transition from simulations to actual deployments can expose inefficiencies, emphasizing the need for models that are robust enough to handle diverse, real-world scenarios [76, 83].
- *Conflict with Traditional Control Systems* (TF = 1; 1 case; 2%): This topic refers to procedural and institutional tensions that arise when AI systems interact with or potentially override established control protocols, decision-making procedures, or regulatory hierarchies. These conflicts are not rooted in hardware limitations but rather in the misalignment between AI-driven autonomy and pre-existing governance mechanisms. For instance, AI may challenge rule-based decision authority by executing actions outside human operator oversight or conflicting with embedded institutional norms [69].

Data governance and security PC (TF = 1, 871; 58 cases; 92%)

This PC addresses key concerns related to data quality, security, privacy, and the potential for monopolization in AI-driven power systems. The safe and effective deployment of AI in this sector requires not only high-quality, standardized data but also robust safeguards to protect data integrity and user privacy. The following topics highlight these challenges:

- *Data Availability and Standardization* (TF = 818; 51 cases; 81%): AI's effectiveness in power systems depends on access to standardized, high-quality data. However, data fragmentation across utilities and power generators, inconsistent regulations,

and the lack of uniform standards create significant barriers. These inconsistencies hinder AI scalability and limit the integration of diverse data sources [18, 84].

- *Data Quality and Bias* (TF = 464; 44 cases; 70%): AI's reliability is directly tied to the quality of the data it learns from. Incomplete or biased datasets can lead to inaccurate, unreliable, or discriminatory AI-driven decisions. In power systems, poor data quality can negatively impact efficiency, sustainability, and equitable resource distribution [75]. Additionally, flawed datasets increase the computational complexity of AI models, reducing their accuracy and effectiveness [7]. Addressing these challenges requires continuous efforts to improve data quality and mitigate bias.
- *Data Privacy and Security* (TF = 190; 17 cases; 27%): The collection, storage, and use of consumer and operational energy data raise significant privacy and security concerns. Beyond individual data rights, this issue extends to protecting data integrity and securing critical infrastructure. As power systems become more AI-dependent, they also become more vulnerable to cyberattacks. Implementing comprehensive cybersecurity measures is crucial to prevent breaches that could disrupt energy operations [1, 14, 82].
- *Cybersecurity Vulnerabilities* (TF = 178; 15 cases; 24%): AI integration introduces new cyber risks to power systems, as advanced automation increases exposure to external threats. The complexity of AI-driven energy management necessitates robust cybersecurity frameworks to protect against attacks that could compromise grid stability and system integrity [14, 85].
- *Consumer Data Ownership* (TF = 162; 18 cases; 31%): This topic focuses on ethical concerns regarding the collection and use of consumer data in energy management. Key issues include data ownership, informed consent, and anonymization practices. Establishing clear and transparent guidelines on these matters is essential to building consumer trust and ensuring that AI-driven energy solutions respect individual privacy [14, 15].
- *Information Monopolization* (TF = 59; 22 cases; 35%): The concentration of data within a few entities can lead to monopolistic control, reducing competition and slowing innovation in the energy sector. As AI systems increasingly rely on vast datasets, the risk of information monopolization grows. To foster a competitive and innovative energy landscape, it is essential to ensure data accessibility and promote open standards [14, 86].

Regulatory and economic concerns: AI as arbitrator

AI is expected to operate within existing energy policy and regulatory structures while balancing concerns of fairness, accountability, and equitable access to its benefits. In this role, AI must navigate the competing interests of governments, regulators, consumers, energy producers, and distributors—each with different perspectives on affordability, access, and sustainability [13, 79]. Additionally, AI's socioeconomic impacts, such as potential job displacement, introduce further complexities to its governance [87]. This MC reflects the tension between AI's ability to drive innovation in the power sector and comply with the regulatory frameworks designed to govern its deployment while safeguarding public welfare. This MC had a term frequency (TF) of 3,831 across 61 cases (97%).

Proponents argue that AI plays a crucial role in enhancing regulatory adaptation and efficiency in the clean energy transition. AI-driven automation can improve fault detection, grid restoration, and policy compliance, enabling regulatory bodies to adapt to rapidly evolving energy technologies. Biagini et al., [5] emphasize how advancements in grid automation will require policymakers to adjust regulatory structures to accommodate new technologies such as autonomous grids:

“Communication technology advancements and higher deployment over all voltage levels can amplify the capability of other grid assets and bring benefit to the system as a whole. We expect that the customer and regulatory pressure on distribution system operators (DSOs) will grow in the future in the direction of maintaining stable, uninterrupted operation (similarly to transmission grid) and data security just to name a few. In this new reality, communication infrastructure will play the vital role.” (p. 5)

However, critics caution that AI-driven policy-making poses significant risks, particularly in areas of accountability and algorithmic bias. Ntoutsis et al., [88] warn that many AI algorithms lack transparency, making it difficult to ensure fairness and neutrality in decision-making:

“Bias is not a new problem rather ‘bias is as old as human civilization’ and ‘it is human nature for members of the dominant majority to be oblivious to the experiences of other groups.’ However, AI-based decision-making may magnify pre-existing biases and evolve new classifications and criteria with huge potential for new types of biases. These constantly increasing concerns have led to a reconsideration of AI-based systems towards new approaches that also address the fairness of their decisions.” (p. 2)

This concern is closely linked to the black box problem and data quality and bias discussed earlier. Without sufficient transparency, stakeholders may find it difficult to trust AI-driven policies, particularly in critical areas such as resource allocation and consumer protection. Another major challenge is the democratization of AI technologies. There is growing concern that AI’s economic benefits may disproportionately favor wealthier regions or corporate entities, exacerbating socioeconomic disparities rather than addressing them. Ahmed and Wahid [89] highlight this risk:

“Our results suggest that rather than the democratization of AI, we observe a concentration of knowledge production by a small group of actors or the de-democratization of AI due to the compute divide.” (p. 5)

Thus, this MC encapsulates the tension between AI’s regulatory potential and the risks of opacity, bias, and inequality in its deployment. Table 5 presents the most common topics of discussion within this MC. The following subsections examine these concerns further through the Points of Contention (PCs) of Accountability and Affordability.

Accountability PC (TF = 3, 158; 60 cases; 95%):

The Accountability PC discusses the regulatory frameworks and evolving policies that influence-and are influenced by-AI adoption and deployment in power systems. As AI

Table 5 Key PCs and topics under regulatory and economic concerns: AI as arbitrator

Regulatory and Economic Concerns: AI as Arbitrator	TF	N Cases	% N Cases
PC: Accountability	3158	60	95
Regulatory compliance	1344	49	78
AI-Specific regulations and supportive policies	1216	56	89
Balancing innovation and protection	342	32	51
Equal distribution of benefits	138	22	35
Impact on Workforce	60	14	22
Understanding AI reasoning	50	10	16
Addressing bias and fairness	8	3	5
PC: Affordability	637	52	83
Development and deployment expense	604	48	76
Return on investment	69	21	33

and data-use regulations continue to co-evolve, ensuring compliance and accountability in AI decision-making is crucial. This PC encompasses several contentious key topics:

- *Regulatory Compliance* (TF = 1, 344; 49 cases; 78%): As regulations surrounding AI evolve, the capacity, especially for power system operators, to interpret and explain AI decisions becomes increasingly vital for ensuring compliance. Hence, in this topic scholars discuss that effective regulatory frameworks will likely require AI systems to provide transparent, auditable decision-making processes that regulators and stakeholders can review [17, 26].
- *AI-Specific Regulations and Supportive Policies* (TF = 1, 216; 56 cases; 89%): This topic discusses the need to update current energy regulations to specifically address AI-related issues, such as algorithmic bias and accountability. Supportive policies, including tax credits and government incentives, are essential for promoting AI adoption while aligning innovation with the public interest [90].
- *Balancing Innovation and Protection* (TF = 342; 32 cases; 51%): Policymakers must strike a balance between fostering AI innovation and protecting consumers from potential risks. This includes addressing liability concerns in cases where AI-controlled grids fail or exhibit unintended biases in resource allocation [14, 16].
- *Equal Distribution of Benefits* (TF = 138; 22 cases; 35%): Ensuring that the benefits of AI-such as increased efficiency and sustainability-are equitably distributed across regions and socioeconomic groups remains a significant challenge. There are concerns that AI-driven improvements might not be universally accessible, possibly deepening current inequalities. A more equitable and just distribution of AI's benefits in power systems is necessary to foster widespread support [19].
- *Impact on Workforce* (TF = 60; 14 cases; 22%): As AI automates tasks within the energy sector, it introduces accountability challenges related to potential job displacement. Policymakers and industry leaders are responsible for managing the social and economic effects of automation, ensuring that the benefits of AI do not come at an undue cost to the workforce. This requires a commitment to support workforce transitions through retraining programs, job placement initiatives, and other support measures that uphold accountability to affected workers and communities [26, 87].
- *Understanding AI Reasoning* (TF = 50; 10 cases; 16%): Transparency in AI decision-making is crucial for accountability. Stakeholders need insight into the decision-

making processes of AI models to build trust and ensure compliance, particularly in areas that affect public services.

- *Addressing Bias and Fairness* (TF = 8; 3 cases; 5%): AI systems must be designed to identify and mitigate biases, ensuring fair decision-making. This is particularly important in public service applications, where bias can lead to unequal access to resources and opportunities [19, 75].

Affordability PC (TF = 637; 52 cases; 83%)

The Affordability PC examines, through a policy lens, the economic and financial debates surrounding the enrollment and mobilization of AI in power systems. High development costs may limit access to wealthier entities, raising concerns about democratization. Topics under this PC include:

- *Development and Deployment Expense* (TF = 604; 48 cases; 76%): The costs of developing and deploying sophisticated AI systems can be prohibitive, potentially limiting access to organizations with substantial financial resources. This raises concerns about the democratization of AI, prompting questions about whether economic policies, such as subsidies or cost-sharing initiatives, could help ensure that AI's benefits are equitably shared across a wider range of stakeholders [26].
- *Return on Investment* (TF = 69; 21 cases; 33%): Decision-makers require clear evidence of AI's cost-benefit ratio to justify investment, particularly in light of economic policies that may support or incentivize AI adoption. Long-term benefits, such as energy savings and reliability gains, must be carefully weighed against the substantial upfront costs of AI development and deployment, with economic frameworks in place to assess and support these investments across diverse energy stakeholders [63, 76].

Interrelatedness between concerns

Figure 1 presents the Co-occurrence Jaccard Coefficient Map, which illustrates the interconnectedness among the four key matters of concern (MCs) identified in our analysis: *Operational Concerns*, *Sustainability Concerns*, *Trust Concerns*, and *Regulatory and Economic Concerns*. The node size reflects each concern's relative term frequency (TF), while the line thickness represents the strength of co-occurrence. Larger nodes indicate higher TF values, and thicker lines correspond to higher Jaccard Coefficients (JCs), as shown by the numbers above the lines.

A *Jaccard Coefficient (JC)* quantifies the degree of co-occurrence between topics, measuring how often they appear together in the same documents rather than just their individual presence. It accounts for both the presence of topics and the relative frequency and patterns of co-occurrence-how extensively two topics are discussed together. For example, suppose **Topic A** (e.g., "AI for renewable energy forecasting") appears in **30 papers**, and **Topic B** (e.g., "AI for grid stability") appears in **40 papers**. However, only **10 papers** discuss both topics together. The **JC** for such a case is calculated as:

$$JC = \frac{|A \cap B|}{|A \cup B|} = \frac{10}{(30 + 40 - 10)} = \frac{10}{60} = 0.17 \quad (1)$$

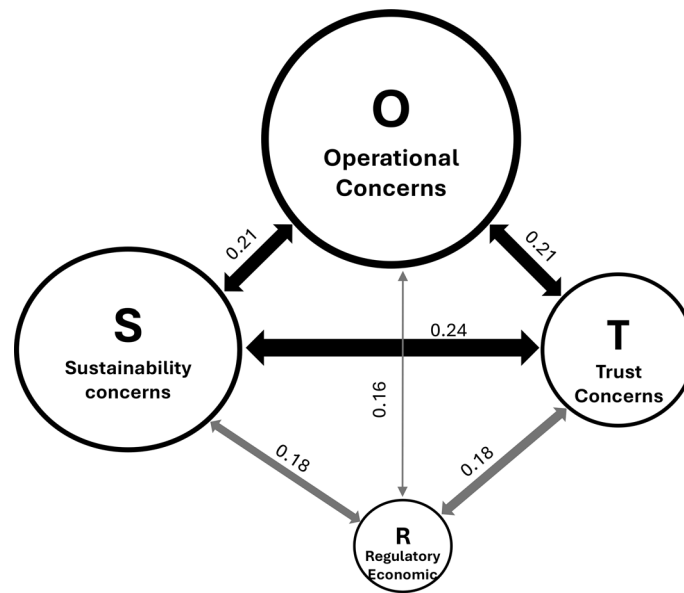


Fig. 1 Co-occurrence Jaccard Coefficient Map. Each MC node size is proportional to its term frequency (TF), and each connecting line's thickness corresponds to its Jaccard Coefficient (JC) value

where $|A \cap B|$ represents the number of documents discussing both topics together, and $|A \cup B|$ represents the total number of documents that mention either topic. A low JC value (~ 0.17) indicates that although both topics appear frequently in the literature, they are rarely explored in direct relation to one another. In contrast, the *N Cases* reported in previous sections simply indicates how many documents reference a topic at least once, without accounting for relationships between topics.

As can be seen, the strongest co-occurrence is observed between *Sustainability Concerns* and *Trust Concerns*, with a JC of **0.24**. This suggests that some discussions on AI's role in promoting energy sustainability intersect with concerns about trust in AI-driven solutions. For example, AI-powered grid optimization models rely on real-time data to enhance energy efficiency, but their effectiveness depends on the transparency of their decision-making processes to gain regulatory and stakeholder acceptance [22]. *Operational Concerns* also exhibit connections with both *Sustainability Concerns* and *Trust Concerns*, each with a JC of **0.21**. This suggests that debates around AI's role in ensuring grid stability, managing distributed energy resources, and maintaining cybersecurity resilience sometimes overlap with sustainability and trust-related issues. For instance, AI is increasingly used to predict and balance fluctuating energy demand, yet its reliability depends on accurate forecasting models and robust cybersecurity safeguards to prevent operational failures [11].

Smaller but relevant connections are found between *Regulatory and Economic Concerns* and both *Sustainability Concerns* and *Trust Concerns*, with JCs of **0.18** each. This indicates that AI discussions in power systems involve considerations related to policy frameworks, economic incentives, and mechanisms to ensure accountability and transparency in AI deployment. For example, government-led incentive programs for AI-driven energy efficiency solutions often require strict guidelines to ensure fair market competition and equitable access to technological advancements [18]. Finally, the co-occurrence between *Regulatory and Economic Concerns* and *Operational Concerns* is **0.16**, reflecting much more limited discussions on aligning AI's operational role with

evolving regulatory landscapes and defining its autonomy within decision-making processes. An example of this is the integration of AI into energy market forecasting, where regulatory bodies must set limits on AI's influence over pricing mechanisms to prevent market distortions [5].

Overall, Figure 1 shows predominantly low JC values, suggesting a fragmented research landscape where most papers focus on a single concern rather than analyzing topics holistically. The combination of low JCs and high individual occurrence rates (*N Cases*) indicates that while many papers mention multiple topics, they typically prioritize topics within the same area concern while only briefly addressing others. This pattern highlights that although AI in power systems intersects with sustainability, trust, and regulatory considerations, it is often approached from a predominantly technical perspective, with limited attention to its broader systemic interactions and interdependencies. These findings underscore the need for a more integrated approach to understanding AI's role in power systems.

Discussion: policy recommendations

This study aimed to explore two core research questions: *What are the primary concerns regarding AI's integration into power systems?* and *What specific contentions emerge from this integration?* The findings presented above indicate that AI's integration into power systems presents challenges that extend beyond isolated technical issues, requiring a comprehensive regulatory and governance approach. Several overarching concerns-such as explainability, cybersecurity, economic accessibility, and sustainability-were prevalent across multiple MCs. To ensure AI's responsible adoption, we propose policy recommendations addressing four critical areas that emerged from the findings: *AI system reliability and integration, explainability and trust, sustainability and energy efficiency, and regulatory alignment with market structures.*

Strengthen AI system reliability and integration standards

AI's ability to enhance energy management depends on its seamless integration into existing power grids while maintaining system stability and interoperability. However, findings highlight concerns regarding AI's black-box nature, legacy infrastructure constraints, and risks of grid destabilization. One major concern is *AI's integrability*-whether it can function effectively within power grids that still rely on outdated control mechanisms and fragmented infrastructure. The study revealed that many legacy systems lack the flexibility to support AI-driven automation, leading to potential bottlenecks in scalability and reliability [58]. Additionally, the lack of standardized integration frameworks makes it difficult for AI systems to operate across diverse grid environments, further complicating large-scale deployment. The findings also emphasize the risks of AI destabilizing power grids, particularly when decision-making is opaque or overly automated. Machlev et al. [22] highlight that AI models with low explainability can introduce unexpected system behaviors, making it difficult for grid operators to diagnose errors and intervene when necessary. To mitigate these risks, energy policies should:

- **Develop interoperability standards** to ensure AI models can integrate with existing and future energy infrastructures, minimizing disruptions in power systems [58].

- **Mandate transparent validation and stress-testing protocols** for AI applications in power grid operations to mitigate risks related to real-time decision-making and system stability [22].
- **Support AI-assisted grid resilience initiatives** by investing in fault-tolerant AI systems that can detect and recover from operational failures [11].

Establish explainability and trust mechanisms for AI decision-making

The opacity of AI systems presents significant challenges in stakeholder trust, cybersecurity, and accountability. Findings suggest that decision-makers and operators struggle to interpret AI-driven recommendations, which raises risks related to system oversight and compliance. One of the most pressing issues is the black-box nature of AI models, particularly those based on deep learning. Many stakeholders in the energy sector express concerns that AI-driven automation may operate without clear human oversight, making it difficult to verify decisions and ensure accountability [22]. Studies indicate that this lack of explainability could erode trust in AI-driven power management, reducing adoption rates among regulators and grid operators. Additionally, concerns around data governance and cybersecurity were widespread in the findings. AI-driven power systems require vast amounts of data, making them highly susceptible to cyber threats. Farhan et al. [82] warn that as AI systems become more deeply embedded in smart grids, they introduce new vulnerabilities that could be exploited, leading to disruptions in power supply and financial losses. To address these challenges, policies should:

- **Require explainability mechanisms** for AI models operating in power systems, ensuring that decision-making processes are auditable and interpretable by regulators and grid operators [80].
- **Implement data governance and cybersecurity standards** to protect AI-driven power systems from cyber threats while ensuring data privacy and consumer protection [82].
- **Introduce accountability frameworks** that define responsibility for AI-induced errors in energy management, including clear liability structures for AI-driven operational failures [18].

Promote AI sustainability and energy efficiency standards

While AI is often promoted as a tool for sustainability, findings indicate concerns about its own energy consumption and environmental impact. The high computational demands of AI models risk offsetting the energy savings they are designed to achieve. One of the most debated concerns in the findings is whether AI truly contributes to long-term sustainability or whether its own energy footprint cancels out its benefits. Ahmad et al. [69] warn that training complex AI models requires significant computational resources, raising ethical and environmental concerns about AI's true contribution to clean energy transitions. Furthermore, the rebound effect was frequently cited in discussions of AI's sustainability. Some studies suggest that while AI can optimize energy use, it may also encourage increased consumption, effectively negating the energy savings it was designed to achieve [70]. This phenomenon underscores the need for regulatory oversight to ensure AI applications align with sustainability targets. In this regard, energy regulations should:

- **Mandate energy efficiency benchmarks** for AI-powered models, ensuring that the energy used in AI computation does not undermine broader sustainability goals [69].
- **Encourage low-energy AI development** through targeted research funding, prioritizing models that minimize computational costs while maintaining accuracy [70].
- **Incentivize AI applications that directly contribute to carbon reduction**, such as AI-driven demand response programs and energy-efficient grid optimization tools.

Align AI integration with market and regulatory Frameworks

Findings suggest that current regulatory frameworks lag behind AI's rapid adoption in power systems, creating gaps in market fairness, consumer protection, and equitable access to AI-driven benefits. One major issue is the economic accessibility of AI technologies. The findings indicate that AI-driven solutions are often monopolized by large corporations, leaving smaller energy providers, prosumers, and developing regions with limited access to AI's benefits [89]. This raises concerns about market fairness and equitable distribution of AI-driven advantages. Additionally, the findings highlight the potential for AI to disrupt energy pricing mechanisms. AI-powered forecasting models can influence electricity prices by dynamically adjusting supply and demand predictions. However, without clear regulatory oversight, AI's involvement in market operations could lead to unfair pricing structures that disadvantage certain stakeholders [5]. To address these regulatory and economic concerns, policies should:

- **Develop AI-specific energy regulations** that address algorithmic bias, economic accessibility, and market fairness, preventing monopolization by large energy corporations [89].
- **Define the scope of AI autonomy in energy markets**, ensuring that AI-driven pricing models and market forecasts do not introduce instability or unfair competition [5].
- **Expand financial incentives for AI adoption** among smaller energy providers, prosumers, and community-driven renewable energy projects to prevent market concentration.

Table 6 summarizes the list of recommendations above. As can be seen, AI presents significant opportunities for power systems but also introduces risks that demand proactive policy intervention. Addressing integration challenges, explainability concerns, sustainability trade-offs, and regulatory gaps is essential for ensuring AI serves as a transparent, efficient, and equitable tool in energy management. These recommendations provide a framework for guiding AI's responsible deployment while maximizing its benefits for energy transition efforts.

Conclusion

This study conducted a systematic review of high-quality academic literature on AI in power systems, employing text analysis to identify key Matters of Concern (MCs) arising from AI integration. The analysis revealed four interconnected MCs (i.e., Operational Concerns, Sustainability Concerns, Trust Concerns, and Regulatory and Economic Concerns) which highlight the complex sociotechnical landscape that the energy sector must navigate to ensure AI's responsible deployment and mobilization.

Table 6 Policy recommendations

Policy Recommendations
Strengthen AI System Reliability and Integration Standards
Develop interoperability standards
Mandate transparent validation and stress-testing protocols
Support AI-assisted grid resilience initiatives
Establish Explainability and Trust Mechanisms for AI Decision-Making
Require explainability mechanisms
Implement data governance and cybersecurity standards
Introduce accountability frameworks
Promote AI Sustainability and Energy Efficiency Standards
Mandate energy efficiency benchmarks
Encourage low-energy AI development
Incentivize AI applications that directly contribute to carbon reduction
Align AI Integration with Market and Regulatory Frameworks
Develop AI-specific energy regulations
Define the scope of AI autonomy in energy markets
Expand financial incentives for AI adoption

Our findings show that AI’s role in power systems extends beyond isolated technical improvements, impacting grid stability, energy sustainability, public trust, and regulatory structures. AI’s potential to enhance forecasting, real-time grid balancing, and energy efficiency is often constrained by legacy infrastructure, explainability challenges, and cybersecurity risks. Additionally, while AI offers tools to support sustainability goals, concerns regarding its own energy consumption and long-term adaptability persist. The lack of transparency in AI-driven decision-making and the absence of well-defined regulatory frameworks further complicate its widespread adoption in power systems.

The primary contribution of this study is the review of the literature in AI in power systems with a matters of concern angle, and development of a comprehensive MC framework, which provides a structured understanding of AI’s interdependencies, risks, and governance needs in power systems. The policy recommendations drawn from our findings emphasize the need to:

1. Strengthen AI system reliability and integration standards, ensuring seamless deployment and interoperability across grids.
2. Establish explainability and trust mechanisms, including transparent AI models and robust data governance frameworks.
3. Promote AI sustainability and energy efficiency standards, addressing the energy-intensive nature of AI models.
4. Align AI deployment with regulatory and market frameworks, ensuring fairness, accountability, and equitable access to AI-driven benefits.

With regard to research limitations, our analysis was restricted to high-quality academic literature, which may not fully capture emerging industry developments, real-world deployment challenges, or broader policy discussions. Given the rapid evolution of AI in power systems, future research should incorporate insights from industry reports, regulatory documents, and case studies to provide a more comprehensive perspective and assess the applicability of the proposed policy recommendations. Additionally, while our computational approach ensures consistency in term identification, it may not fully capture implicit or contextual connections between discussions, potentially overlooking

nuances in how topics are framed within specific debates. AI-assisted dictionary construction, despite expert validation, has inherent limitations in capturing the full complexity of discourse. Future research could employ more sophisticated qualitative discourse analysis methods to better identify latent relationships and provide a deeper understanding of how AI-related concerns in power systems are interconnected.

Finally, in terms of impact, our research shows that AI's integration into power systems represents a transformative shift that requires balancing innovation, ethical considerations, and regulatory oversight. Our research highlights the importance of an inclusive, cross-sectoral approach to ensure AI supports a sustainable, equitable, and resilient energy future. We hope this study contributes to the ongoing debate by promoting a systems-oriented and comprehensive perspective on AI integration in power systems. By fostering collaboration among policymakers, industry stakeholders, and researchers, we aim to encourage AI-driven advancements that align with broader energy transition goals while safeguarding public trust, system reliability, and environmental responsibility.

Supplementary Information

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Supplementary file 1

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Author contributions

Felipe Henao: Conceptualization; Formal analysis; Supervision; Investigation; Writing - original draft, review & editing. Robert Edgell: Conceptualization; Formal Analysis; Methodology; Validation; Supervision; Writing - original draft, review & editing; Visualization. Ambar Sharma: Data Curation; Methodology; Investigation; Writing - original draft. Jeffrey Olney: Formal analysis; Software; Methodology.

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Data availability

Data and the full corpus will be made available on request.

Declarations

Consent to publish

Not applicable.

Ethics declaration

Not applicable.

Ethics, consent to participate, and consent to publish

Not applicable.

Competing interests

The authors declare no competing interests.

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